

TRIGONOMETRY — UNIT 2

Polytechnic Engineering Mathematics - Easy Notes

Unit-2 notes trigonometric ratios, identities, standard values basic applications exam-oriented revision

1. Trigonometric Ratios

- $\sin \theta = \frac{\text{Perpendicular}}{\text{Hypotenuse}}$
- $\cos \theta = \frac{\text{Base}}{\text{Hypotenuse}}$
- $\tan \theta = \frac{\text{Perpendicular}}{\text{Base}}$
- $\operatorname{cosec} \theta = 1/\sin \theta$, $\sec \theta = 1/\cos \theta$, $\cot \theta = 1/\tan \theta$

2. Fundamental Relations

- $\tan \theta = \sin \theta / \cos \theta$
- $\cot \theta = \cos \theta / \sin \theta$
- $\sin^2 \theta + \cos^2 \theta = 1$
- $1 + \tan^2 \theta = \sec^2 \theta$
- $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$

3. Standard Values

- 0° : $\sin=0$, $\cos=1$, $\tan=0$
- 30° : $\sin=1/2$, $\cos=\sqrt{3}/2$, $\tan=1/\sqrt{3}$
- 45° : $\sin=1/\sqrt{2}$, $\cos=1/\sqrt{2}$, $\tan=1$
- 60° : $\sin=\sqrt{3}/2$, $\cos=1/2$, $\tan=\sqrt{3}$
- 90° : $\sin=1$, $\cos=0$, $\tan=\text{undefined}$

4. Complementary Angles

- $\sin(90^\circ - \theta) = \cos \theta$
- $\cos(90^\circ - \theta) = \sin \theta$
- $\tan(90^\circ - \theta) = \cot \theta$
- $\cot(90^\circ - \theta) = \tan \theta$
- $\sec(90^\circ - \theta) = \operatorname{cosec} \theta$

5. Allied Angles - Basic

- $\sin(-\theta) = -\sin \theta$
- $\cos(-\theta) = \cos \theta$
- $\tan(-\theta) = -\tan \theta$
- $\sin(180^\circ - \theta) = \sin \theta$

- $\cos(180^\circ - \theta) = -\cos \theta$
- $\tan(180^\circ - \theta) = -\tan \theta$

6. Important Identities - Practice

- $\sin \theta = 3/5$, $\cos \theta = 4/5$ (acute angle)
- $\tan \theta = 3/4$, $\sec \theta = 5/4$
- $1 - \sin^2 \theta = \cos^2 \theta$ $1 - \cos^2 \theta = \sin^2 \theta$

7. Heights and Distances - Basic

When given unknown height/distance problems, use $\tan$, $\sin$ or $\cos$ to solve for the unknown side.

Example: Given distance d and angle of elevation θ , find height $h = d \tan \theta$.

8. Important Exam Questions

1. $\sin$, $\cos$ or $\tan$ to solve for unknown side
2. Reciprocal trigonometric ratios
3. $\sin^2 \theta + \cos^2 \theta = 1$ identity
4. Given $\sin \theta = 5/13$, find $\cos \theta$ or $\tan \theta$
5. 30° , 45° or 60° standard values
6. Complementary angles identities
7. Given $\tan \theta = 3/4$, find $\sin \theta$ or $\cos \theta$
8. $1 + \tan^2 \theta = \sec^2 \theta$ identity
9. Height and distance using angle of elevation

9. Quick Revision Table

Angle	sin	cos	tan
0°	0	1	0
30°	$1/2$	$\sqrt{3}/2$	$1/\sqrt{3}$
45°	$1/\sqrt{2}$	$1/\sqrt{2}$	1
60°	$\sqrt{3}/2$	$1/2$	$\sqrt{3}$
90°	1	0	Undefined

Exam Tip: Standard values and 3 fundamental identities are essential; practice numerical questions.